Q12

According to Kinetic Theory, the average KE of an ideal gas atom is $K = \frac{3}{2}kT$.

Mass of a single Neon atom:

$$m = \frac{\text{molar mass}}{\text{Avogadro's constant}} = \frac{20.2 \text{ x } 10^{-3}}{6.02 \text{ x } 10^{23}} \text{ kg}$$

$$\frac{1}{2}m{v_{rms}}^2 = \frac{3}{2}kT$$

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

$$= \sqrt{\frac{3(1.38\text{x}10^{-23})(30+273)}{\frac{20.2 \text{ x } 10^{-3}}{6.02 \text{ x } 10^{23}}}} = 612 \approx 610 \text{ m s}^{-1}$$

Ans: C

Q13